FLATNESS IS THE SAME OVER THE BASE AND OVER AN ETALE COVER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- An etale morphism is flat, and flatness is preserved by composition. Morphisms, label morphisms-lemma-etale-flat (proof line 139); Algebra, label algebra-lemma-composition-flat (proof line 141). Gives the easy implication from flatness over the etale scheme to flatness over the base.
- Base change of a flat module, and the diagonal of an etale morphism is an open immersion. Morphisms, labels morphisms-lemma-base-change-module-flat (proof line 147), morphisms-lemma-diagonal-unramified-morphism and morphisms-lemma-etale-smooth-unramified (proof lines 150-151). The first makes the pullback of the module to the product flat over the etale scheme, and the others identify the source with an open subscheme of the product on which the restriction of the second projection is the given morphism.

Source reference. Original lemma 00HC: sources/stacks-chow/tags.txt, source lines 638–638. Navigation pointer for \ref{algebra-lemma-composition-flat} used inside the counted span (line 141); the argument that consumes it is retained in the span.

Source reference. Original lemma 01U8: sources/stacks-chow/tags.txt, source lines 2401–2401. Navigation pointer for \ref{morphisms-lemma-base-change-module-flat} used inside the counted span (line 147); the argument that consumes it is retained in the span.

Source reference. Original lemma 02GE: sources/stacks-chow/tags.txt, source lines 3225–3225. Navigation pointer for \ref{morphisms-lemma-diagonal-unramified-morphism} used inside the counted span (line 150); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 02GS: sources/stacks-chow/tags.txt, source lines 3239–3239. Navigation pointer for \ref{morphisms-lemma-etale-flat} used inside the counted span (line 139); the argument that consumes it is retained in the span.

Source reference. Original lemma 02GK: sources/stacks-chow/tags.txt, source lines 3231–3231. Navigation pointer for \ref{morphisms-lemma-etale-smooth-unramified} used inside the counted span (line 151); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $X \to T \to S$ be morphisms of schemes with $T \to S$ étale. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $x \in X$ be a point. Then*

$$\mathcal{F} \text{ flat over } S \text{ at } x \Leftrightarrow \mathcal{F} \text{ flat over } T \text{ at } x$$

In particular $\mathcal{F}$ is flat over S if and only if $\mathcal{F}$ is flat over T.

Proof. As an étale morphism is a flat morphism (see Morphisms, Lemma 02GS) the implication "$\Leftarrow$" follows from Algebra, Lemma 00HC. For the converse assume that $\mathcal{F}$ is flat at x over S. Denote $\tilde{x} \in X \times_S T$ the point lying over x in X and over the image of x in T in T. Then $(X \times_S T \to X)^*\mathcal{F}$ is flat at $\tilde{x}$ over T via $\mathrm{pr}_2 : X \times_S T \to T$, see Morphisms, Lemma 01U8. The diagonal $\Delta_{T/S} : T \to T \times_S T$ is an open immersion; combine Morphisms, Lemmas 02GE and 02GK. So X is identified with open subscheme of $X \times_S T$, the restriction of pr_2 to this open is the given morphism $X \to T$, the point $\tilde{x}$ corresponds to the point x in this open, and $(X \times_S T \to X)^*\mathcal{F}$ restricted to this open is $\mathcal{F}$. Whence we see that $\mathcal{F}$ is flat at x over T. □